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**Annasaheb Dange College of Engineering and
Technology (ADCET), Ashta**

Department of CSE (IOT and CSBT)
Quality Laboratory Manual
Introduction to
Cyber Security Lab
2ICPC205

Second Year

Sem-IV

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Course Instructor
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Experiment No. 2

Title of Experiment: Design and Simulate a Basic Network Topology Using Network Tools.

Packet Tracer.

Aim of Experiment: To design and simulate a Local Area Network (LAN) consisting of one switch and twenty PCs using Cisco Packet Tracer, configure IP addresses, and verify communication among all devices.

Course Outcomes (CO)

CO1: Understand the basic concepts of computer networks and network devices.

CO2: Design and simulate a simple LAN using Cisco Packet Tracer.

CO3: Verify network connectivity using simulation tools.

System Requirements:

-Cisco Packet Tracer (Version 8.2 or above)

-Computer with Windows/Linux Operating System

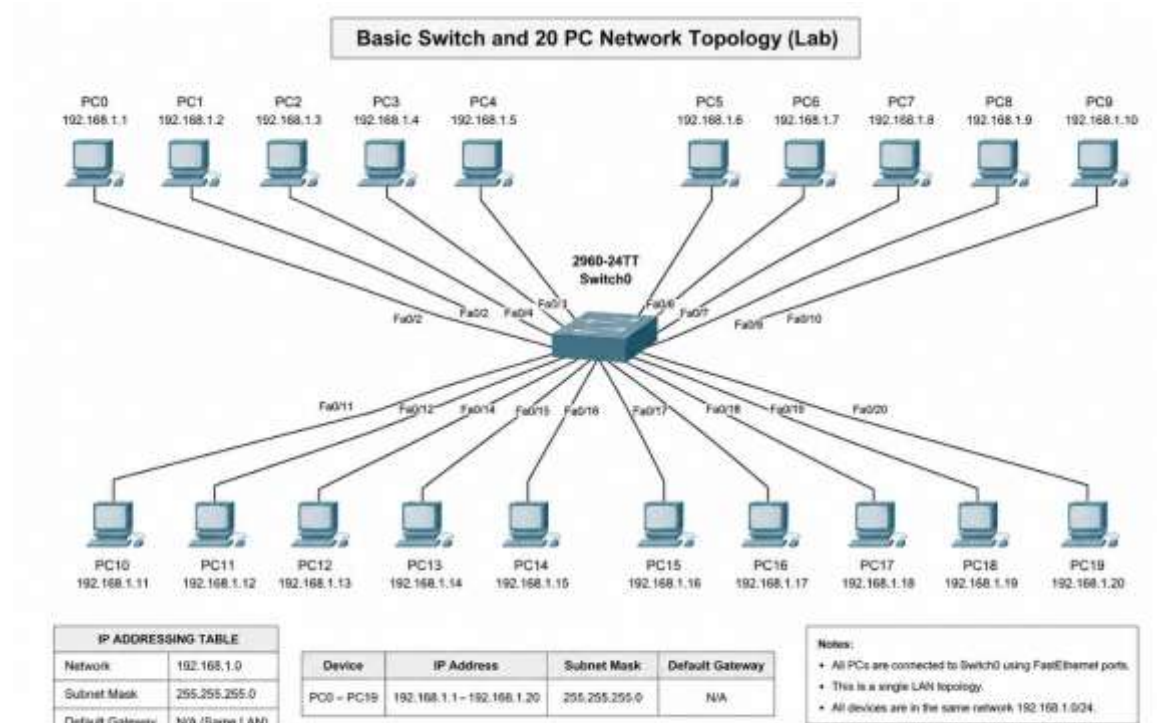
Theory:-

A **Local Area Network (LAN)** is a network that interconnects devices within a limited geographical area such as a computer laboratory, office, or campus. The most commonly used LAN topology is the **Star Topology**, where every device is connected to a central switch.

A **switch** operates at the Data Link Layer (Layer 2) of the OSI model. It forwards frames based on MAC addresses, allowing efficient communication between connected devices. In this experiment, all 20 computers belong to the same network, so they can communicate directly through the switch without requiring a router.

Network Topology

- 1 Cisco 2960-24TT Switch
- 20 PCs
- Copper Straight-Through cables
- Network: 192.168.1.0/24



IP Addressing Table

Device	IP Address	Subnet Mask
PC0	192.168.1.1	255.255.255.0
PC1	192.168.1.2	255.255.255.0
PC2	192.168.1.3	255.255.255.0
PC3	192.168.1.4	255.255.255.0
PC4	192.168.1.5	255.255.255.0
PC5	192.168.1.6	255.255.255.0
PC6	192.168.1.7	255.255.255.0
PC7	192.168.1.8	255.255.255.0
PC8	192.168.1.9	255.255.255.0
PC9	192.168.1.10	255.255.255.0
PC10	192.168.1.11	255.255.255.0
PC11	192.168.1.12	255.255.255.0
PC12	192.168.1.13	255.255.255.0
PC13	192.168.1.14	255.255.255.0

Device IP Address Subnet Mask

PC14 192.168.1.15 255.255.255.0

PC15 192.168.1.16 255.255.255.0

PC16 192.168.1.17 255.255.255.0

PC17 192.168.1.18 255.255.255.0

PC18 192.168.1.19 255.255.255.0

PC19 192.168.1.20 255.255.255.0

Default Gateway: Not Required (Single LAN)

Procedure

Step 1: Create the Topology

1. Open Cisco Packet Tracer.
2. Place one Cisco 2960-24TT Switch.
3. Add 20 PCs.
4. Connect each PC to the switch using Copper Straight-Through cables.

Step 2: Configure IP Addresses

For each PC:

1. Click PC → Desktop → IP Configuration.
2. Assign the IP address according to the table.
3. Set the subnet mask to 255.255.255.0.
4. Leave the default gateway blank.

Step 3: Save the Topology

Save the Packet Tracer file as:

Exp02_StarTopology.pkt

Step 4: Verify Network Connectivity

Open Command Prompt on PC0 and execute:

ping 192.168.1.20

Similarly test:

ping 192.168.1.10

ping 192.168.1.15

ping 192.168.1.18

Expected Output:

Reply from 192.168.1.20

Packets: Sent = 4

Received = 4

Lost = 0 (0% loss)

Step 5: Simulation

1. Switch to Simulation Mode.
2. Select Add Simple PDU.
3. Click PC0 as the source.

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4. Click PC19 as the destination.
 5. Observe packet transmission through the switch.
 6. Repeat for different PC pairs.

Observations

Activity	Expected Result	Actual Result
PC0 → PC5	Ping Successful	_____
PC0 → PC10	Ping Successful	_____
PC0 → PC19	Ping Successful	_____
PC8 → PC15	Ping Successful	_____
Packet Delivery	Successful	_____

Result

A star topology consisting of one switch and twenty PCs was successfully designed and simulated in Cisco Packet Tracer. All devices communicated successfully using IP addressing within the same subnet, demonstrating proper LAN configuration.

Learning Outcomes

After completing this experiment, students will be able to:

- Design a star topology in Cisco Packet Tracer.
- Connect multiple hosts using a Layer-2 switch.
- Configure IPv4 addresses on end devices.
- Test network connectivity using the **ping** command.
- Observe Ethernet frame forwarding in Simulation Mode.
- Understand communication within a single broadcast domain.

Conclusion :-**Expected Oral Questions –**

Q. No.	Viva Question	CO Mapped
1	What is a LAN?	CO1
2	What is a star topology?	CO1
3	Why is a switch used in a LAN?	CO1
4	At which OSI layer does a switch operate?	CO1
5	What is a MAC address?	CO1

6	What is an IP address?	CO1
7	Why is the subnet mask required?	CO1
8	Why is a router not required in this experiment?	CO1
9	Which cable is used to connect a PC to a switch?	CO2
10	What does the ping command verify?	CO3
11	What happens if two PCs are assigned the same IP address?	CO3

Interview Questions:-

- | | | |
|----|---|-----|
| 1. | What is a broadcast domain? | CO1 |
| 2. | How many ports does a Cisco 2960-24TT switch provide? | CO2 |
| 3. | What is the purpose of Simulation Mode in Packet Tracer? | CO3 |
| 4. | What are the advantages of a star topology over a bus topology? | CO1 |